#HW 1

Graded

Student

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Total Points

100 / 100 pts

Question 1

Q1

20 / 20 pts

- **– 2 pts** There is an error in your calculation. Feet should be converted to inches. ft^3 can be converted to inch^3 by multiplying it with 12^3.

✔ **– 0 pts** Click here to replace this description.

- **– 2 pts** Since the Patm is given in lbf so you should get your gauge and system pressure in lbf too. So you should divide rhogh by the gravity.
- **– 2 pts** Absolute or system pressure can be obtained by adding up the gauge pressure with Patm.
- **– 2 pts** The effective height should be 15*sin 40 degrees. You would have got a different answer if you had used 40 degree instead of 30 degree since the question mentioned a 40 degree inclination.
- **– 5 pts** Late

Question 2

Q2

20 / 20 pts

✔ **– 0 pts** Click here to replace this description.

- **– 4 pts** Since, the specific volume has been given, in order to get the density, you need to take the reciprocal of the specific volume.
- **– 1 pt** The answer you got is in Pascal. To convert it to kPa you must divide it by 1000.
- **– 5 pts** Incomplete submission
- **– 5 pts** Late

Question 3

Q3 **20** / 20 pts

✔ **− 0 pts** Click here to replace this description.

- **− 2 pts** In the question it has been mentioned that the pressure changes from 1 bar to 200 bar in increments and volume should be calculated next. You had increased volume and then calculated pressure. You are supposed to increase the pressure.

- **− 2 pts** Pressure should be on the y axis and volume should be on the x axis

- **− 2 pts** Please remember to share your data from the next time. The question also mentioned that you should share the screenshot of both the plot and data.

- **− 2 pts** The question said to add 20 increments. So you should have reported 20 data sets.

- **− 2 pts** Please remember to add titles on your axes from the next time.

- **− 10 pts** Incomplete

- **− 5 pts** Late

- **− 1 pt** Add units on axes.

Question 4

Q4 **15** / 15 pts

✔ **− 0 pts** Click here to replace this description.

- **− 2 pts** The temperature change is not the same for all. It is the same for C and K. It is the same for F and R. You should clearly state it.

- **− 2 pts** The conversion scale for F to C is wrong. It is (F-32)/1.8 i.e. the entire (F-32) gets multiplied by 5/9 or divided by 9/5.

- **− 5 pts** Incomplete

- **− 1 pt** You should show the conversion of both 70 degree F and 42 degree F. You have not shown conversion for the 70 degree F.

- **− 5 pts** Late

Question 5

Q5 **25** / 25 pts

✔ **− 0 pts** Click here to replace this description.

- **− 2 pts** You are supposed to calculate the absolute pressure in 5a. So you add the Patm too for getting that.

- **− 1 pt** Pressure inside the submersible is same as the atmospheric pressure.

- **− 1 pt** You should deduct Patm from Pabs to get gauge pressure

- **− 10 pts** Incomplete

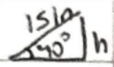
- **− 10 pts** Late

Questions assigned to the following page: 2, 3, 4, and 1

CHE 201 09/05/25

Homework 1

Question 1:



(a) $15 \cdot \sin 40° = 9.64\ in = 0.803\ ft$ $\quad \Delta p = \rho g h = \rho h$ $\quad (0.803\ ft)(54.2\ lbf/ft^3) = 43.5\ lbf/ft^2$

$\sin 40 = \frac{h}{15}$ $\quad \Delta P = \frac{43.5\ lbf/ft^2}{144} = 0.302\ lbf/in^2$ $\quad P_{gas} = P_{atm} + \Delta P = 14.7\ lbf/in^2 + 0.3\ lbf/in^2 = \boxed{15 \frac{lbf}{in^2}}$

(b) $P_{gage} = P_{gas} - P_{atm} = 15.0\ lbf/in^2 - 14.7\ lbf/in^2 = \boxed{0.302\ psi\ \text{so gage}}$

(c) An inclined manometer has greater sensitivity and accuracy for measuring small pressure differences because the inclined tube spreads the liquid level change over a longer distance, making it easier to read.

Question 2:

(a) $v = 0.00122\ m^3/kg$ $\quad \Delta p = p_a - p_b = \rho g L$

$\rho = 1/v = 819.7\ kg/m^3$ $\quad = (819.7\ kg/m^3)(9.81\ m/s^2)(0.12\ m)$

$L = 12\ cm = 0.12\ m$ $\quad = 964.9\ Pa = \boxed{0.965\ kPa}$

(b) The pressure decreases as kerosene flows from point a to point b as pipe diameter decreases.

Question 3: (uploaded Excel picture)

Question 4:

(a) 42°F:

$T(°C) = \frac{5}{9}(42-32) = \boxed{5.56°C}$

$T(K) = 5.56°C + 273.15 = \boxed{278.7K}$

$T(°R) = 42°F + 459.67 = \boxed{501.67°R}$

70°F:

$T°C = \frac{5}{9}(70-32) = \boxed{21.11°C}$

$T(K) = 21.11°C + 273.15 = \boxed{294.3K}$

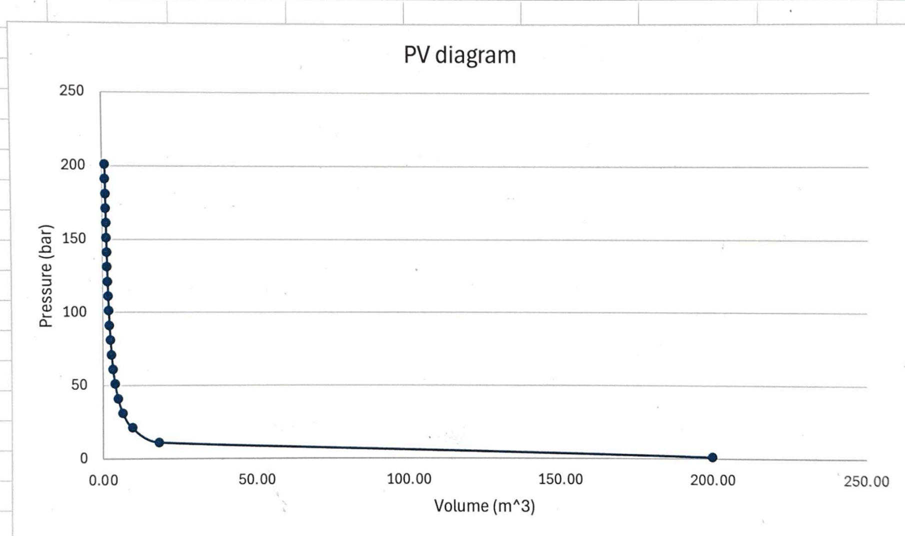
$T(°R) = 70°F + 459.67 = \boxed{529.67°R}$

(b) $\Delta T = 70°F - 42°F = \boxed{28°F}$ $\quad \Delta T(°C) = 21.11°C - 5.56°C = \boxed{15.6°C}$

$\Delta T(°R) = \boxed{28°R}$ $\quad \Delta T(K) = \boxed{15.6K}$

Question assigned to the following page: 3

Volume (m^3)	Pressure (bar)
200.00	1
18.18	11
9.52	21
6.45	31
4.88	41
3.92	51
3.28	61
2.82	71
2.47	81
2.20	91
1.98	101
1.80	111
1.65	121
1.53	131
1.42	141
1.32	151
1.24	161
1.17	171
1.10	181
1.05	191
1.00	201



Questions assigned to the following page: 4 and 5

(c) A change of $1°F = 1°R$; °F and °R give same change
(d) A change of $1°C = 1\,K$; °C and K give same change

Question 5:
(a) $p = p_{atm} + \rho g h = 101{,}325\ Pa + (1{,}025\ kg/m^3)(9.81\ m/s^2)(3{,}800\ m) = 3.84 \times 10^7\ Pa$
MPa: $\frac{3.84\times10^7}{10^6} = \boxed{38.4\ MPa}$
atm: $\frac{3.84\times10^7}{101325} = \boxed{378\ atm}$
psi: $\frac{3.84\times10^7}{6895} = \boxed{5556\ psi}$
(b) $p_{gauge} = \rho g h = (1025\ kg/m^3)(9.81\ m/s^2)(3{,}800\ m) = \boxed{38.2\ MPa}$
Gauge pressure matters because structures feel pressure relative to atmosphere, not the absolute total including 1 atm.
(c) The sub is kept near 1 atm. Outside ~38 MPa vs inside ~0.1 MPa is big pressure difference
(d) A submersible withstands the extremely high hydrostatic pressures of deep ocean through strong hull design (titanium/thick steel sphere) that resists compression; spherical shape distributes stress uniformly.
(e) The titan submersible imploded because its carbon-fiber titanium hull could not withstand the immense hydrostatic pressure at 3,800 m, leading to sudden structural failure.
The key engineering failures that contributed to disaster were the use of materials unsuited for compressive deep-ocean loads, repeated fatigue without proper certification, and inadequate external safety oversight.